Natural Language Processing Analysis of Financial Markets Using News, Themes, and Earnings Signals

A Structured Study on Signal Interpretation, Context, and Market Behaviour

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1. Project Overview

This project builds on the limitation observed in a previous machine learning approach, where structured data alone was not sufficient to explain market behaviour. Increasing the complexity of the model did not improve understanding, which led to a shift in focus from model-based analysis to data-based exploration.

In reality, market movement is not driven only by numerical data. It is also influenced by how information is perceived by market participants. This information exists in the form of unstructured data such as financial news, global economic events, and company earnings communication. These are not directly measurable like price data, but they play an important role in shaping market behaviour.

The objective of this project is to understand whether such unstructured data can provide meaningful signals that help explain market movement. Instead of focusing on prediction, the approach is centred on extracting different types of signals from text and analysing how they align with actual market outcomes.

The project follows a structured, question-driven approach. Each type of signal is analysed step by step, starting from simple representations like sentiment, then moving to more complex structures like entities and themes, and finally combining them to understand their overall effectiveness. The aim is not to prove that one signal works perfectly, but to understand which type of information carries meaningful insight and where the limitations exist.

Overall, this project shifts the focus from model performance to signal understanding, where the goal is to interpret how information flows through the market rather than trying to directly predict its movement.

2. Problem Statement

In the previous machine learning project, the focus was on using structured market data to understand index-level behaviour. However, one key limitation observed was that increasing the complexity of the model did not lead to better understanding of the dataset or improvement in meaningful insights. This raised an important question about whether the limitation lies in the model itself or in the type of data being used.

In reality, market behaviour is not driven only by numerical data. It is part of a broader cause-and-effect chain, where the first layer is how market participants perceive information. This perception is formed through unstructured data such as financial news, macroeconomic developments, and company earnings communication. These elements influence decisions, which then reflect in price movements.

Based on this understanding, the core problem of this project is to examine whether incorporating unstructured data can provide additional insight into market behaviour. Instead of focusing on improving model performance, the project shifts towards evaluating whether signals extracted from text can meaningfully align with or explain market movement.

Another important challenge in this problem is related to data availability and quality. The dataset spans a long period, but the availability of relevant news and earnings information is limited and uneven. This creates constraints in analysis and also introduces potential bias based on how the data is collected.

Overall, the problem is not just about testing a new type of data, but about understanding whether unstructured information can act as a meaningful layer in explaining market behaviour, and what limitations arise when working with such data.

3. Dataset and Project Setup

This project is built using a combination of unstructured text data and structured market data. The unstructured side consists of financial news and earnings-related information, while the structured side includes market returns for the Nifty 50 index and selected stocks.

The news dataset is not continuous in nature. Since complete historical news data is difficult to obtain over a long period, the dataset focuses on selected and relevant news items that capture major events and important developments. This makes the analysis event-based, where only specific days with meaningful news are considered, rather than treating every day equally.

Alongside this, the market dataset contains daily return values for the Nifty 50 index and selected stocks. These returns represent the actual market outcome and are used as the reference point against which all text-based signals are evaluated. The structure of the data is maintained in a way that allows alignment through the Date column, so that signals extracted from news or earnings can be directly compared with market movement on the same day.

The workflow of the project is divided into two main stages. The first stage involves extracting signals from text data using Python, where NLP techniques such as sentiment analysis, entity recognition, and phrase extraction are applied. The second stage involves structuring and analysing this data in BigQuery SQL, where views are created to organise each step of the analysis and to maintain a consistent pipeline.

The project follows a view-based architecture rather than relying on static tables. Each step is saved as a separate view, which acts as a checkpoint. This makes the system modular and flexible, allowing new data to be added without disturbing the existing structure. It also helps in validating each part of the analysis independently.

One important limitation in the dataset is related to availability and bias. Since the data spans a long time period, the number of news and earnings observations is limited and uneven. This introduces constraints in the analysis and also reflects how unstructured data is more difficult to collect and standardise compared to structured market data.

Project Setup Summary

- **Data Type:** Unstructured text (news, earnings) + structured market returns
- **Analsyis Unit:** Event-based (focused on news days, not full daily coverage)
- **Signal Definition:** Text-derived signals (sentiment, entity, theme, earnings)
- **Output Variable:** Market return (Nifty index and selected stocks)
- **Processing Flow:** Python (NLP extraction) → BigQuery SQL (data structuring and analysis)
- **System Design:** View-based modular pipeline for flexibility and validation

4. Analytical Path / Project Method

The project is designed as a step-by-step process, where each stage addresses a specific question and the outcome of that stage leads to the next area of exploration. The focus is not on applying different NLP techniques independently, but on understanding what each signal contributes to the overall picture.

The analysis begins with the most direct signal that can be extracted from text, which is sentiment. The initial question is whether the tone of financial news itself carries any meaningful information about market movement. This serves as a basic way to convert unstructured text into a measurable form.

After testing sentiment, a limitation becomes evident. Sentiment captures the tone of the news, but it does not reflect the actual content or subject being discussed. This leads to the next stage, where the focus shifts toward entities. Instead of focusing only on how the news feels, the analysis moves toward understanding what the news is about and whether those underlying subjects show any consistent relationship with market behaviour.

However, even with entity-based analysis, some challenges remain. Individual entities, even when important, often appear inconsistent or noisy when analysed in isolation. This leads to a broader shift in approach. Rather than focusing on individual elements, the analysis moves toward themes, where multiple related signals are grouped into a single narrative. This is based on the idea that market behaviour may respond more to a combination of related factors than to isolated pieces of information.

The results from theme-based analysis show relatively stronger patterns, which raises another question. If different signals capture different aspects of the same information, combining them should ideally provide a more complete signal. This leads to the multi-signal stage, where sentiment, entity, and theme are combined. However, instead of improving the results, this introduces additional noise, indicating that combining weak or partially useful signals does not necessarily lead to a stronger outcome.

At this stage, the analysis shifts toward a different type of signal, which is earnings data. Unlike news, earnings reflect the company's own communication and perspective. The idea is to evaluate whether combining this company-level signal with broader theme-based signals can create a more meaningful structure.

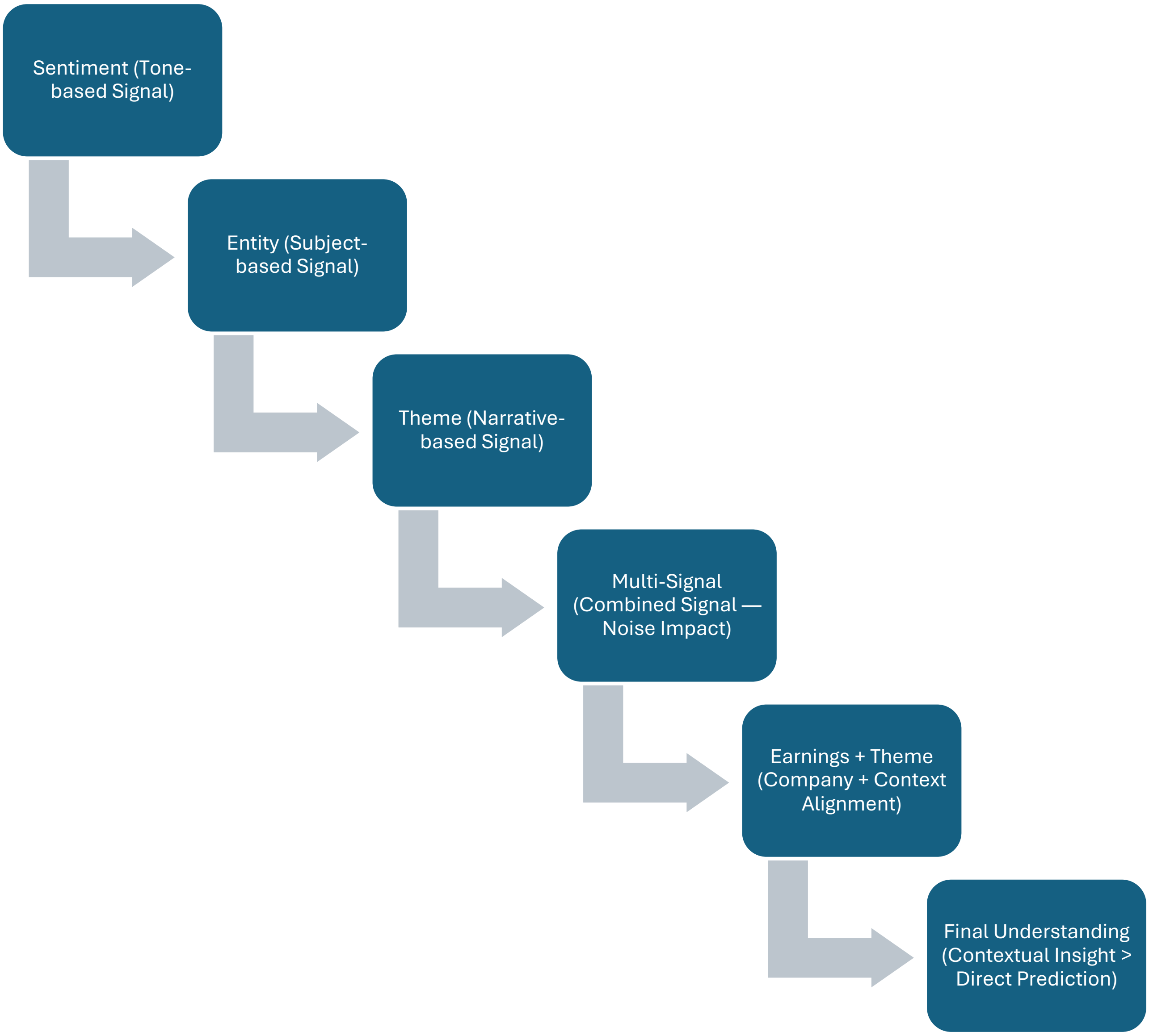
The final stage focuses on understanding whether this combination improves alignment with market behaviour, particularly at the stock level. While some improvement is observed in company-specific cases, the results remain limited at the overall market level.

Overall, the project progresses as a sequence of questions rather than a sequence of techniques. Each stage is driven by a limitation observed in the previous step, and the approach evolves based on what the data is able to show, rather than following a predefined structure.

Step-wise Signal Flow of the Project

- **Stage 1:** Sentiment analysis to test whether tone alone provides a meaningful signal
- **Stage 2:** Entity analysis to understand whether underlying subjects improve the signal
- **Stage 3:** Theme analysis to capture broader narratives and reduce noise
- **Stage 4:** Multi-signal combination to test whether signals complement or dilute each other
- **Stage 5:** Earnings + theme analysis to evaluate company-level and contextual alignment

Figure 1. Signal Evolution Flow of the Project



5. Major Question 1 — Can sentiment from financial news act as a meaningful signal for market behaviour?

5.1 The Question

The first major question of this project begins with the most direct way of converting text into a measurable signal, which is sentiment. The core question here is simple: can the tone of financial news provide any meaningful signal about market behaviour?

Since financial news reflects ongoing events, expectations, and reactions in the market, it is natural to think that the overall tone of the news may align with how the market moves. If this relationship exists, sentiment can act as a basic signal for understanding market behaviour.

5.2 Why This Stage Was Needed

This stage acts as the starting point for the entire NLP analysis. Sentiment is one of the simplest ways to convert unstructured text into numerical form. It reduces complex text into a directional signal, which makes it easier to compare with market data.

Before moving into more detailed structures like entities or themes, it is important to first check whether this basic signal itself carries any useful information. If sentiment does not show any meaningful relationship, then relying on more complex NLP structures without understanding this limitation may not be useful.

5.3 Method Used

To extract sentiment from financial news, I used NLP-based sentiment analysis to convert text into numerical values.

Each news item was processed to generate two main outputs:

- **Sentiment Score (Polarity)** → represents the direction of the tone
- **Magnitude Score (Strength)** → represents how strongly the tone is expressed

The sentiment score ranges between -1 and +1:

- Positive values indicate positive tone
- Negative values indicate negative tone
- Values near zero indicate neutral tone

The magnitude score captures the intensity of the sentiment, regardless of direction.

Since BigQuery SQL does not directly support NLP functions, the sentiment extraction was performed using Python. The Vader sentiment analyser was used, which is a rule-based model that assigns scores based on predefined word patterns.

From Vader:

- The **compound score** was used as the sentiment score
- The **sum of positive and negative scores** was used as the magnitude

After extracting these values, the data was moved into BigQuery and aggregated at the date level. This step was necessary because multiple news items can exist on the same day, while market data is available as one value per day.

- Average sentiment → used to capture overall daily tone
- Sum of magnitude → used to capture total intensity of news on that day

This created a uniform dataset where each date had a single sentiment signal, which could then be aligned with market returns.

5.4 Key Findings

After aligning the sentiment signal with market returns, the results did not show a strong or consistent relationship. The overall alignment between sentiment direction and market movement was around 50%, which is close to a random outcome.

There were many instances where:

- Positive sentiment did not lead to positive market movement
- Negative sentiment did not lead to negative market movement

This mismatch occurred frequently and indicated that sentiment alone is not a reliable signal for understanding market behaviour.

Another observation was that even when sentiment was positive, the strength of that sentiment did not consistently translate into stronger market movement. This suggests that tone alone may not capture the full context required to explain market reactions.

5.5 Interpretation

For me, this stage was important not because sentiment was expected to give a strong result, but because it helped establish a baseline. The results clearly showed that sentiment, when used alone, behaves like a weak signal.

One key reason for this is that sentiment captures only the tone of the news, not the underlying content or importance of the information. A news item can have a positive tone but may not be relevant enough to influence market behaviour.

Another important point is that market movement depends on multiple factors, including expectations, macro conditions, and existing market positioning. Sentiment alone is not able to capture these complexities.

So while sentiment provides a measurable signal, it does not consistently translate into actual market movement.

5.6 Transition

This stage helped clarify one important limitation. If sentiment captures only the tone but not the actual subject or importance of the news, then the next step is to shift the focus toward what the news is about rather than how it feels.

That is what leads to the next question: if the analysis moves from sentiment to entities, which represent the key subjects of the news, does the signal become more meaningful?

6. Major Question 2 — Can entities from financial news provide a more meaningful signal than sentiment?

6.1 The Question

After observing that sentiment alone does not provide a strong or consistent signal, the next question naturally becomes more focused on the content of the news. The key question here is: instead of looking at how the news feels, can focusing on what the news is about provide a more meaningful signal for understanding market behaviour?

Entities represent the main subjects of the news, such as companies, events, or macroeconomic factors. The idea is that these underlying subjects may have a more direct connection with market movement compared to sentiment, which only captures tone.

6.2 Why This Stage Was Needed

This stage becomes important because of the limitation observed in sentiment analysis. Sentiment reduces the entire text into a single directional signal, but it ignores the actual drivers behind the news. A news item can have a positive tone but may not be relevant enough to influence the market.

So the focus shifts toward identifying the key elements within the news that may actually drive market behaviour.

Entities act as those elements. They help in identifying:

- which companies are being discussed
- which events are taking place
- which macroeconomic factors are being highlighted

The expectation here is that by focusing on these core subjects, the signal may become more meaningful and less dependent on tone.

6.3 Method Used

To extract entities from the news data, I used Named Entity Recognition (NER) through the spaCy NLP model in Python.

NER identifies important elements in a sentence and classifies them into categories such as:

- organisations (ORG)
- locations (GPE)
- and other entity types

However, extracting entities alone is not sufficient. Not all entities are equally important for analysis.

To address this, a salience or importance score was created for each entity based on two factors:

- **Entity Type Score** → certain types like companies were given higher importance
- **Position Score** → entities appearing earlier in the sentence were considered more important

The final salience score was calculated by combining these two factors. After extracting and scoring the entities, the data was moved into BigQuery SQL for further analysis.

To identify which entities matter more, two measures were used:

- frequency of occurrence
- average salience score

These were combined to create an overall importance measure for each entity. Based on this, only the more relevant entities were selected for further analysis.

6.4 Key Findings

After filtering for important entities and aligning them with market returns, the results did not show a strong or consistent relationship.

Some entities appeared frequently, but their relationship with market returns was not stable. For example:

- certain company-related entities showed both positive and negative returns
- some high-frequency terms did not carry meaningful importance

A few event-based entities showed relatively stronger signals, but these were limited in number and not consistent across the dataset.

Overall, most entities did not demonstrate a clear or reliable pattern when compared with market movement.

6.5 Interpretation

From my analysis, focusing on entities did not significantly improve the signal compared to sentiment. Even after applying importance filtering, many entities behaved like noise when analysed individually.

One key reason for this is that entities represent isolated components of the news, but market behaviour is influenced by a combination of factors rather than a single element. Another important observation is related to data limitations. The dataset is limited and selective, which means the frequency and coverage of entities are not uniform. This affects the stability of the results.

Also, the process of selecting and scoring entities involves some level of subjectivity, especially in assigning importance weights. This further impacts the reliability of the signal.

Overall, entity-based analysis provides some level of contextual understanding, but it does not act as a strong standalone signal for market behaviour.

6.6 Transition

This stage highlights another limitation. If both sentiment and entities, when analysed individually, do not provide strong or consistent signals, then focusing on isolated elements may not be the right approach.

This leads to the next question: instead of analysing individual signals, can grouping related patterns into broader themes provide a more meaningful understanding of market behaviour?

7. Major Question 3 — Can themes provide a stronger and more meaningful signal for market behaviour?

7.1 The Question

After observing that both sentiment and entity-based analysis do not provide strong or consistent signals, the next question becomes broader. If analysing individual elements of the news is not sufficient, can grouping related patterns into themes provide a more meaningful understanding of market behaviour?

The idea here is that market movement may not react to isolated signals, but rather to a combination of related factors that together form a broader narrative.

7.2 Why This Stage Was Needed

This stage comes from a limitation observed in the previous steps. Sentiment captures only tone, and entity analysis focuses on individual subjects, but both approaches treat information in a fragmented way. However, market behaviour is usually influenced by multiple factors working together.

So instead of looking at single signals, the focus shifts toward identifying patterns that represent a broader context. Themes act as that broader structure. They combine multiple related phrases or patterns into a single idea, which can better reflect how information is perceived in the market.

This also helps reduce some of the bias seen in entity-based analysis, as themes are less dependent on specific names and more focused on overall situations or events.

7.3 Method Used

To construct themes, the first step was to extract meaningful phrases from the news data. Using NLP techniques in Python, specifically noun chunk extraction through spaCy, the text was broken into smaller meaningful units. These phrases represent what is happening in the news rather than just identifying who is involved.

Since the initial output contained a large number of phrases, many of which appeared only once or twice, filtering was required.

- Only phrases with frequency greater than or equal to 3 were retained
- This helped remove noise and focus on more consistent patterns

After filtering, the remaining phrases were grouped manually into themes. Each theme represents a broader idea formed by combining related phrases. For example:

- phrases related to growth, expansion, and performance were grouped together
- phrases related to risk, uncertainty, or disruption were grouped into another theme

This grouping process is subjective in nature, as it depends on how the phrases are interpreted and classified. Once the themes were created, they were aligned with market returns to observe whether any consistent relationship exists between these themes and market behaviour.

7.4 Key Findings

Compared to sentiment and entity-based analysis, themes showed relatively stronger and more interpretable patterns.

Some themes demonstrated a clearer relationship with market movement:

- Themes related to positive market sentiment, such as investor confidence or strong performance, showed positive alignment with market returns
- Themes related to risk, uncertainty, or disruptions showed negative alignment

These results were also more intuitive, as they matched how market behaviour is generally understood in real-world situations.

However, not all themes showed consistent patterns. Some themes remained weak or inconsistent, indicating that not every grouped pattern translates into a meaningful signal.

7.5 Interpretation

From my analysis, themes provided a more meaningful way to understand market behaviour compared to sentiment and entities.

One key reason is that themes capture a broader narrative rather than focusing on isolated elements. This makes the signal more aligned with how market participants interpret information. Another important point is that themes help reduce some of the noise seen in entity-based analysis. Instead of focusing on individual entities, the analysis focuses on overall situations or developments.

However, this approach also comes with limitations. The process of creating themes is subjective, and the results depend on how phrases are grouped. Different grouping methods may lead to different outcomes. Also, data scarcity remains an issue. Since the dataset is limited, the number of observations for each theme is not large, which affects the stability of the results.

Overall, themes emerge as a relatively stronger signal, but they are not completely reliable on their own.

7.6 Transition

This stage brings an important insight. Themes provide a better understanding compared to individual signals, but they are still not fully consistent.

This leads to the next question: if sentiment, entities, and themes all capture different aspects of the same information, can combining these signals create a more complete and stronger representation of market behaviour?

8. Major Question 4 — Does combining multiple NLP signals create a stronger signal for market behaviour?

8.1 The Question

After analysing sentiment, entities, and themes separately, one natural question comes up. If each of these signals captures a different aspect of the same information, can combining them create a stronger and more complete signal for understanding market behaviour?

The expectation here is that while individual signals may be weak or partial, combining them may improve the overall quality of the signal.

8.2 Why This Stage Was Needed

This stage comes from the mixed results observed in earlier steps.

- Sentiment was weak
- Entity analysis remained inconsistent
- Theme analysis showed relatively stronger signals

Since themes capture broader narratives and sentiment captures tone, combining them may help add context. Similarly, entities represent specific elements that may add more detail.

So, the idea is that combining these signals could help capture multiple dimensions of the same information. However, there is also a possibility that weaker signals may introduce noise and reduce the effectiveness of stronger signals. This stage is meant to test which of these outcomes actually happens.

8.3 Method Used

To combine the signals, a unified scoring approach was created. Each signal was first converted into a structured numerical form:

- **Sentiment Direction** → +1 (positive), -1 (negative), 0 (neutral)
- **Magnitude** → used as the strength of sentiment
- **Theme Strength** → categorised as high, medium, or low
- **Entity Strength** → categorised as high or low

To create a combined signal, these values were multiplied to generate a single score for each observation. Since multiple signals could exist for the same date, the combined scores were aggregated at the date level using an average.

To evaluate alignment with market behaviour:

- The combined score was converted into directional form (+1, -1, 0)
- This was compared with the direction of market returns
- Matches and mismatches were calculated

Additionally, the combined scores were grouped into categories such as high, medium, and low strength to check whether stronger signals show better alignment.

8.4 Key Findings

The results from the combined signal did not show improvement. In fact, they were weaker than expected.

- The mismatch rate was higher than the match rate
- Overall alignment was worse than sentiment alone
- Even stronger signal categories did not show significant improvement

There was a slight reduction in mismatch when moving from low to high signal strength, but this improvement was not strong enough to be considered meaningful. Another important observation was that the number of strong signals was very limited, which made it difficult to draw reliable conclusions.

8.5 Interpretation

From my analysis, combining multiple signals did not lead to a stronger outcome. Instead, it introduced additional noise into the system.

One key reason is that not all signals carry equal strength. When weak signals like sentiment and entity are combined with a stronger signal like themes, they can dilute the overall effect.

Another important factor is the method used to combine signals. Using simple multiplication and averaging may not be sufficient to capture the complexity of how these signals interact. There is also a level of subjectivity in assigning weights and combining signals, which further affects the reliability of the results.

Overall, this stage shows that combining multiple signals does not automatically improve the quality of analysis. In some cases, it can reduce clarity by introducing noise.

8.6 Transition

This stage leads to an important conclusion. If combining multiple weak or partially useful signals does not improve the outcome, then a different approach may be needed. This brings the focus to a new type of signal, which is earnings data. Unlike news-based signals, earnings reflect the company's own perspective and communication.

The next question is whether combining a relatively stronger signal like themes with earnings data can produce a more meaningful and stable signal.

9. Major Question 5 — Can earnings data and themes together provide a stronger signal for market behaviour?

9.1 The Question

After observing that combining sentiment, entity, and theme signals did not improve the results, the focus shifts to a different type of signal. The question here is whether earnings data, which reflects the company's own communication, can provide a meaningful signal, and whether combining it with theme-based signals can improve the overall understanding of market behaviour.

The idea is that earnings represent the company's perspective, while themes reflect the broader market context. If both align, the signal may become stronger and more meaningful.

9.2 Why This Stage Was Needed

This stage comes from two observations in the earlier analysis. First, theme-based signals showed relatively better results compared to sentiment and entity analysis. Second, combining multiple weak signals did not work and instead introduced noise.

So instead of combining all signals, the focus shifts to combining only two signals:

- one relatively strong signal (themes)
- one different type of signal (earnings data)

Earnings data is important because it reflects how companies present their performance, expectations, and outlook. This can influence how investors interpret company-specific developments.

However, earnings communication also carries a natural bias, as companies tend to present information in a positive way. This becomes an important factor in analysing its effectiveness.

9.3 Method Used

The first step was to extract sentiment from earnings text data using NLP techniques. Using Python and the Vader sentiment analyser:

- The **compound score** was used as the sentiment (direction)
- The **sum of positive and negative scores** was used as the magnitude (strength)

These values were aggregated at the company and date level to create a single earnings signal for each earnings event. For the combined signal:

- Earnings sentiment was categorised based on strength
- Theme signals were classified based on their average return impact

These two were then combined into categories such as:

- strong positive (both signals aligned positively)
- conflicting (signals not aligned)
- weak (both signals weak or unclear)

The combined signal was then compared with:

- stock-level returns
- market-level returns

The combined signal was then compared with both stock-level returns and overall market returns to evaluate whether the signals align consistently with actual market movement.

9.4 Key Findings

The earnings signal on its own did not show strong or consistent alignment with market behaviour. One clear observation was that most earnings sentiment values were positive, which reflects the natural bias in company communication.

When earnings signals were combined with themes:

- A large portion of cases fell into the strong positive category
- Some conflicting cases were observed, showing differences between company communication and broader market context

At the stock level, the combined signal showed relatively better alignment, reaching around 70% in some cases.

However, at the market level, the results remained close to 50%, similar to previous observations. Another limitation was the small number of usable earnings data points, as many earnings announcements occurred on non-trading days, reducing the available sample size.

9.5 Interpretation

From my analysis, earnings data provides some additional insight, but it is not strong enough on its own to explain market behaviour. One key limitation is the positive bias in earnings communication. Companies tend to present their performance in a favourable way, which reduces the neutrality of the signal.

When combined with themes, there is some improvement at the stock level. This suggests that when company-level communication aligns with broader market context, the signal becomes more meaningful.

However, this improvement does not extend to the overall market level. The results remain close to a random outcome, which has been a consistent pattern throughout the project. Another important factor is data limitation. The number of earnings observations is small, which affects the reliability of conclusions.

Overall, this stage shows that combining a stronger signal (themes) with a different type of signal (earnings) can provide some improvement at a specific level, but it is not sufficient to explain broader market behaviour.

9.6 Final Understanding

This stage completes the overall structure of the project. Across all stages, one pattern remains consistent:

- individual signals are weak
- combining signals is not straightforward
- broader narrative-based signals perform relatively better

Themes provide the strongest understanding among all signals, while earnings data adds some value at the company level when aligned with those themes.

However, none of the signals are strong enough to consistently explain market behaviour at the index level.

10. Overall Interpretation of the Project

This project started with a simple idea of understanding whether unstructured text data such as news and earnings can explain market behaviour in a meaningful way. As the analysis progressed through different stages, one clear pattern started to emerge.

The initial approach, which focused on sentiment, did not provide strong results. While sentiment captures the tone of the news, it does not capture the actual drivers behind it. This made it a weak signal for understanding market movement.

The next stage, entity-based analysis, shifted the focus toward identifying what the news is about. However, even after applying importance filtering, most entities did not show a stable or consistent relationship with market returns. This indicated that analysing individual elements in isolation does not provide a reliable signal.A more meaningful improvement was observed when the analysis moved to themes. By grouping related phrases into broader narratives, the signal became more interpretable and aligned better with market behaviour. This suggests that market movement is better understood at a narrative level rather than through isolated signals.

However, when different signals were combined together, the results did not improve. Instead, the combined signal introduced additional noise, which reduced the effectiveness of the stronger theme signal. This shows that combining multiple signals does not automatically create a better outcome, especially when the underlying signals are not equally strong.

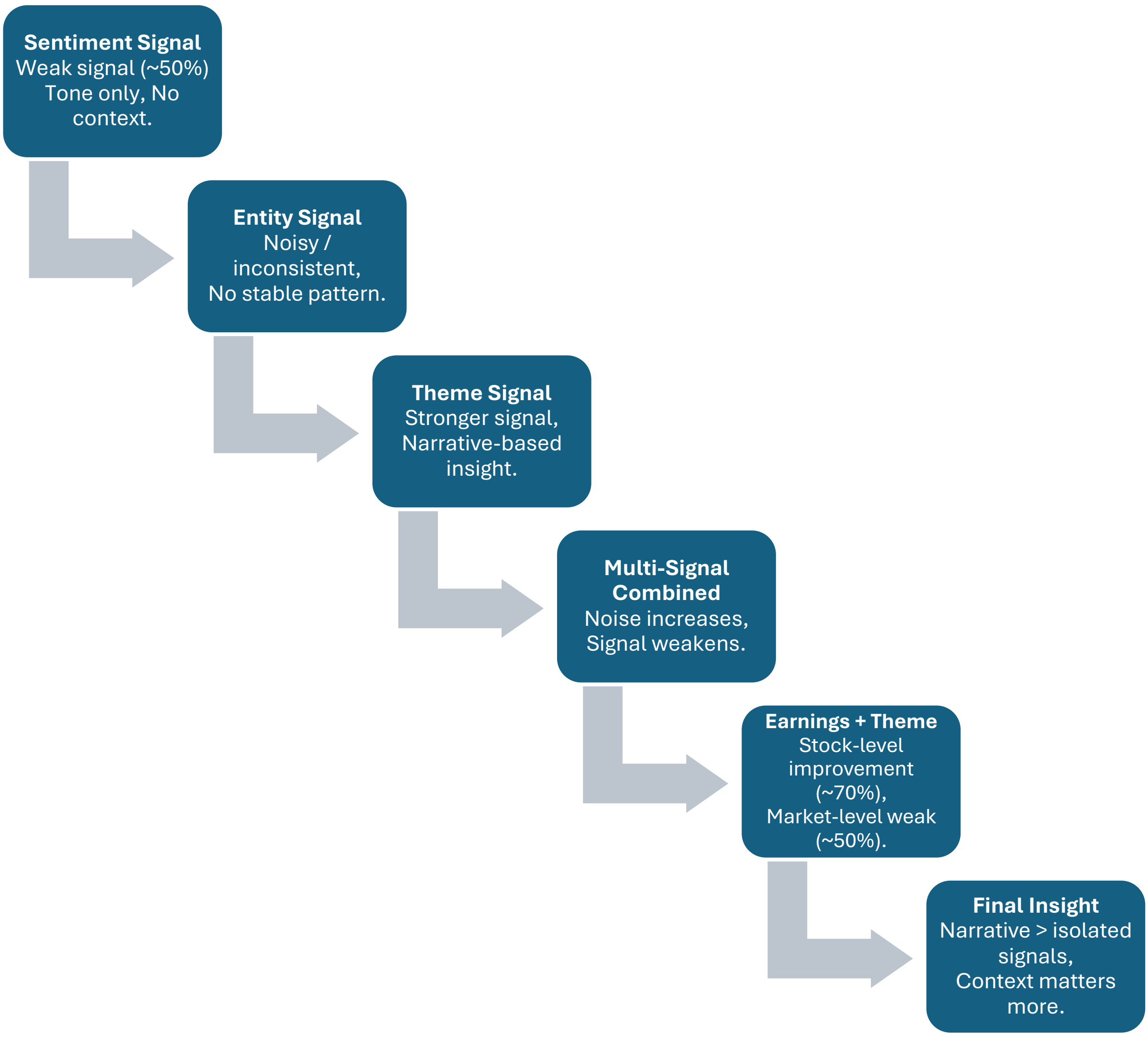
In the final stage, earnings data was introduced as a different type of signal representing the company’s own perspective. When combined with themes, some improvement was observed at the stock level, indicating that alignment between company communication and broader market context can provide additional insight. However, this improvement did not extend to the overall market level, where the results remained close to a random outcome.

Overall, the project shows that unstructured data can provide useful context, but not all extracted signals are equally meaningful. Among all the approaches, theme-based analysis stands out as the most effective way to interpret market behaviour, while other signals either remain weak or introduce noise when combined without a strong structure.

The progression of these signals and their relative effectiveness, as shown earlier in the signal flow diagram, reflects how the project evolved from simple to more structured forms of analysis.

The overall effectiveness of each signal and how the understanding evolved across the project is summarised in the figure below.

Figure 2. Summary of Signal Effectiveness Across the Project



11. Key Learnings from the Project

This project was not only about testing different NLP techniques, but also about understanding how unstructured data behaves when used to interpret market movement. As the analysis progressed, a few important learnings became clear.

The first learning is that unstructured text data is inherently subjective. Unlike structured market data, where values are fixed and measurable, text data depends on how it is collected, filtered, and interpreted. This makes the outcome sensitive to the methodology used, and different approaches can lead to different results.

Another important learning is that not all extracted signals are equally meaningful. Sentiment and entity-based signals, even after applying filters and scoring methods, did not show consistent relationships with market behaviour. This indicates that extracting information alone is not enough, and the way that information is structured plays a critical role.

A key shift in understanding came with theme-based analysis. Grouping related patterns into broader narratives provided a more meaningful way to interpret the data. This suggests that market behaviour is better understood through context and combined effects rather than through isolated elements.

The project also highlighted that combining multiple signals is not straightforward. The expectation was that combining sentiment, entities, and themes would create a stronger signal. However, the results showed that weaker signals can introduce noise and reduce the effectiveness of stronger ones. This emphasises the importance of signal design and weighting, rather than simply increasing the number of inputs.

Another important learning comes from earnings data. While it represents a direct communication from companies, it carries a natural positive bias. This affects the neutrality of the signal and limits its ability to explain market behaviour independently. However, when aligned with broader themes, it can provide some additional insight at the company level.

One of the most important overall learnings is that unstructured data is more useful for providing context than for direct prediction. The signals extracted from text help in understanding why certain movements may happen, but they do not consistently indicate what will happen next.

In summary, this project shows that working with unstructured data requires careful interpretation, awareness of bias, and a focus on structure rather than just extraction. The value of NLP in this context lies more in enhancing understanding than in building direct predictive signals.

12. Limitations of the Project

This project involves working with unstructured text data, which brings certain limitations that affect the overall strength and reliability of the results. These limitations are not only related to the data itself, but also to how the data is collected, processed, and interpreted.

One of the main limitations is the scarcity of data. Since the dataset spans a long time period, collecting consistent and relevant news and earnings data for every day is difficult. As a result, the analysis becomes event-based rather than continuous, which reduces the number of usable observations and affects the stability of the findings.

Another important limitation is the bias in data collection. News data is not uniformly distributed, and the selection of news can be influenced by the availability of sources or focus on certain companies or events. This introduces bias into the dataset, which can affect the signals extracted from it.

Earnings data also carries an inherent limitation in the form of positive bias. Companies generally present their performance in a favourable way, which leads to sentiment scores being skewed towards the positive side. This reduces the neutrality of the earnings signal and limits its effectiveness when used independently.

The process of theme creation introduces subjectivity into the analysis. Since themes are formed through manual grouping of phrases, the results depend on how the grouping is done. Different interpretations of the same phrases could lead to different themes and different outcomes.

There is also a limitation related to data alignment. Many news and earnings events occur on non-trading days such as weekends, where corresponding market returns are not available. These observations have to be excluded, which further reduces the dataset and may lead to loss of relevant information.

Finally, there are tool-related limitations. While BigQuery SQL is effective for structured data analysis, it does not directly support advanced NLP functions. This requires additional processing in Python and introduces dependency on manual logic, which can affect consistency and scalability.

The key limitations of the project are summarised below:

Table 1. Summary of Key Limitations

Limitation	Impact
Data scarcity	Reduced sample size and weaker statistical reliability
News collection bias	Skewed representation of events and entities
Earnings bias	Overrepresentation of positive sentiment
Manual theme grouping	Subjectivity in signal construction
Non-trading day mismatch	Loss of observations due to missing market data
Tool limitations	Dependence on external processing and manual logic

13. Future Improvements

Based on the limitations observed during the project, there are several areas where the analysis can be improved. These improvements are not only about adding more data or complexity, but also about refining how unstructured text is processed and interpreted.

One of the most important improvements is related to data availability. Using a shorter and more recent time period with higher news frequency can help create a more consistent dataset. This would reduce the issue of data scarcity and improve the reliability of the results.

Another area of improvement is in the way themes are constructed. In this project, themes were created through manual grouping, which introduces subjectivity. Using more systematic or automated approaches for grouping phrases could help make the theme structure more consistent and less dependent on individual interpretation.

Signal design is another important aspect. Instead of using simple combinations like multiplication or averaging, more refined methods can be explored to better capture the interaction between different signals. This could help reduce the noise observed in the multi-signal analysis.

Earnings data can also be improved by increasing the number of observations and aligning them more accurately with market trading days. This would help in better understanding the relationship between earnings communication and market movement.

Another potential improvement is in how signals are aligned with market behaviour. Instead of comparing signals only with same-day returns, exploring lag-based relationships (such as next-day or short-term impact) could provide a more realistic understanding of how the market reacts to information.

Finally, using tools that provide direct NLP support within the analysis environment could help reduce dependency on manual processing and improve scalability. This would make the workflow more efficient and easier to extend.

The key areas for improvement are summarised below:

Table 2. Potential Areas for Improvement

Area	**Improvement**
Data coverage	**Use shorter and more recent periods with higher news frequency**
Theme construction	**Apply more systematic or automated grouping methods**
Signal design	**Develop better weighting and combination approaches**
Earnings data	**Increase observations and improve date alignment**
Signal alignment	**Explore lag-based relationships with market returns**
Tool capability	**Use platforms with better NLP integration**

14. Final Answer to the Problem Statement

The main objective of this project was to understand whether unstructured data such as financial news, themes, and earnings information can provide meaningful insight into market behaviour beyond traditional structured data.

From the overall analysis, one clear understanding emerges. Unstructured data does provide additional context, but not all signals extracted from it are equally useful for explaining market movement.

Sentiment, which captures only the tone of the news, did not show a strong or consistent relationship with market returns. The results remained close to a random level, which indicates that tone alone is not sufficient to explain market behaviour.

Entity-based analysis improved the focus by identifying what the news is about, but most entities did not show stable or consistent patterns. Even after applying importance filtering, entities often behaved like noise when analysed individually.

The most meaningful improvement came from theme-based analysis. By grouping related patterns into broader narratives, themes provided clearer and more interpretable signals. This suggests that market behaviour is better understood at a contextual level rather than through isolated elements.

However, combining multiple signals did not improve the results. Instead, it introduced additional noise and reduced the effectiveness of the stronger signals. This shows that simply combining different signals does not guarantee a better outcome, especially when the signals are not equally strong.

Earnings data added another dimension by representing the company's perspective. While it showed some improvement at the stock level when combined with themes, it remained limited in explaining broader market behaviour. The presence of positive bias in earnings communication also affected its reliability.

Across all stages, one consistent observation remains. The alignment with overall market behaviour stays around a mid-level outcome, indicating that market movement is influenced by many factors that are not fully captured by the dataset used in this project.

Overall, the project shows that unstructured data is more useful for providing context rather than for direct prediction. Theme-based signals offer the most meaningful insight, while other signals either remain weak or introduce noise when not structured properly.

In conclusion, unstructured data can enhance the understanding of market behaviour, but it does not provide a strong or consistent standalone explanation for market movement. Its real value lies in supporting interpretation rather than replacing structured analysis. This reinforces that understanding market behaviour requires combining multiple dimensions beyond what limited unstructured data alone can capture.

15. Appendix / Note on Supporting Assets

The main report presented here is based on a detailed working document developed during the actual execution of the NLP project. This underlying document captures the step-by-step analysis, observations, and interpretation as the signals were extracted, structured, and evaluated in real time.

This working document is included in the project repository as the **NLP Project Live Working Analysis** file. It reflects the complete thought process followed during the project, including how each stage evolved from one question to the next.

In addition to this, the repository also contains the **NLP Output Fact Sheet**, which provides structured summaries of key results and observations used throughout the analysis.

Supporting assets such as processed datasets, intermediate views, and relevant outputs used during the analysis are also included. These files help in understanding how the unstructured text data was transformed into structured signals and how the evaluation was carried out.

Overall, these supporting materials provide a complete view of both the analytical process and the underlying data used in the project.